

Energy and Backreaction Bounds for Finite-Pointer Observer Entropy

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Abstract

The observer rule of Harlow, Usatyuk, and Zhao is controlled by a second Rényi entropy, but does not derive the field resources needed to produce its pointer-basis record. We give such a bound in an exactly solvable local field model. A finite pointer controls compact sources for a free scalar, producing a Schur dephasing channel with conditional final data p_i . If w_i are the pointer weights, $P_{\text{cl}} = \sum_i w_i^2$, and $\bar{E}_K = \sum_i w_i \|p_i - \bar{p}\|^2 / 2$, then final support in $[0, L]$ implies

$$\text{Tr } \rho_P^2 \geq P_{\text{cl}} + (1 - P_{\text{cl}}) \exp \left[-\frac{C_\beta(L) \bar{E}_K}{1 - P_{\text{cl}}} \right], \quad S_2(\rho_P) \leq \min\{H_2(w), C_\beta(L) \bar{E}_K\}.$$

The coefficient is exact: $C_\beta(L) = 2L\Lambda(\pi L/\beta)$, where $\Lambda(\tau)$ is the simple top eigenvalue of the positive compact kernel $\pi^{-1} \log[\sinh(\tau(u+v))/\sinh(\tau|u-v|)]$ on $(0, 1)$. The entropy bound is saturated in the symmetric binary top-mode sector; no general- d sharpness is claimed. For the conformally coupled massless scalar in a four-dimensional de Sitter static patch, $\beta = 2\pi R$, the same coefficient controls all angular and canonical sectors, with $C_{2\pi R}(L) = RC_{\text{opt}}(L/R)$.

Using a purified field record as the nonideal observer record in the simple Harlow–Usatyuk–Zhao random code, the Haar-averaged squared inner-product fluctuation for an orthogonal CRT-real matter pair is therefore bounded below by the displayed physical purity, up to the exact finite-code factor $D/(D+2)$. Finally, if every conditional spherical $q_i = 0$ branch obeys a local final-slice constraint budget $Q_{b,i} \leq \delta$, then

$$S_2(\rho_P) \leq \delta \frac{R^2}{G} \frac{C_{\text{opt}}(y) \tanh y \operatorname{sech}^2 y}{2}.$$

This is a fixed-background channel plus a branchwise final-slice constraint corollary, not a coupled gravity theorem. The prescribed pointer and source exclude actuator work, autonomous-observer stress, and a deterministic error floor for every fixed code.

1 Introduction

The observer rule of Harlow, Usatyuk, and Zhao replaces the inner product of closed-universe states by a quantum-to-classical channel carried by an observer record [1]. In their simple random-code model, departures from an ideal inner product are controlled by the purity $e^{-S_2(\omega_{\text{ob}})}$ of that record. This makes the second Rényi entropy operational, but leaves a complementary physical question: how much localized field energy is required to produce a record with a specified purity?

We answer that question in an exactly solvable model. A finite pointer controls compact classical sources for a free scalar field in a thermal static patch. The exact time-ordered evolution is a controlled Weyl

displacement, so tracing the field gives a Schur dephasing channel. If w_i is the initial pointer distribution and p_i are its conditional post-switch momentum data, define

$$P_{\text{cl}} = \sum_i w_i^2, \quad \bar{p} = \sum_i w_i p_i, \quad \bar{E}_K = \frac{1}{2} \sum_i w_i \|p_i - \bar{p}\|^2. \quad (1)$$

Here P_{cl} is the purity of the ideal classical distribution and $\bar{E}_K$ is the field energy that distinguishes the conditional records; a common displacement costs energy but carries no pointer information.

Suppose every conditional datum is supported in the half-line interval $[0, L]$. Our central finite-pointer result is

$$\text{Tr } \rho_P^2 \geq P_{\text{cl}} + (1 - P_{\text{cl}}) \exp \left[-\frac{C_\beta(L) \bar{E}_K}{1 - P_{\text{cl}}} \right], \quad (2)$$

$$S_2(\rho_P) \leq \min\{H_2(w), C_\beta(L) \bar{E}_K\}. \quad (3)$$

The continuous value is understood when $P_{\text{cl}} = 1$. The coefficient is the exact localized norm of the thermal inverse Hamiltonian,

$$C_\beta(L) = 2L\Lambda(\pi L/\beta), \quad k_\tau(u, v) = \frac{1}{\pi} \log \frac{\sinh[\tau(u+v)]}{\sinh[\tau|u-v|]}, \quad (4)$$

where $\Lambda(\tau)$ is the simple top eigenvalue of the positive compact operator with kernel k_τ on $L^2(0, 1)$. The purity bound follows by combining the sharp pairwise form estimate with a weighted variance identity and Jensen's inequality. Equal binary weights and opposite top-eigenfunction profiles saturate it. We do not claim global sharpness for arbitrary pointer dimension.

The operator theorem underlying equation (4) is independent of de Sitter geometry. For Dirichlet half-line momentum data,

$$\langle p, h_0^{-1} \coth(\beta h_0/2) p \rangle \leq \frac{1}{2} C_\beta(L) \|p\|^2, \quad (5)$$

and the exact coefficient obeys the global bracket

$$\max \left\{ \frac{3L}{\pi}, \frac{16L^2}{\beta\pi^2} \right\} \leq C_\beta(L) \leq \frac{4 \text{arsinh}(1)}{\pi} L + \frac{16L^2}{\beta\pi^2}. \quad (6)$$

Uniform remainders identify the vacuum-controlled small-support law and the thermal large-support law.

We then specialize to the conformally coupled massless scalar in a four-dimensional de Sitter static patch. The Bunch–Davies temperature has $\beta = 2\pi R$, and the optical partial-wave operators are

$$A_\ell = -\partial_x^2 + \frac{\ell(\ell+1)}{R^2 \sinh^2(x/R)}. \quad (7)$$

Resolvent order shows that no higher angular momentum has a larger thermal momentum form. A separate localized estimate controls the field-coordinate sector. Consequently, the half-line momentum optimizer is the unique optimum over the full angular and canonical phase space, with

$$C_{2\pi R}(L) = R C_{\text{opt}}(y), \quad C_{\text{opt}}(y) = 2y\Lambda(y/2), \quad y = L/R. \quad (8)$$

The physical record also fits directly into the simple observer code. Purifying the conditional field states gives

$$|\omega\rangle = \sum_i \sqrt{w_i} |i\rangle_{\text{Ob}} |e_i\rangle_{\text{Ob}'}, \quad e^{-S_2(\omega_{\text{Ob}'})} = \text{Tr } \rho_P^2. \quad (9)$$

For an orthogonal CRT-real matter pair, the exact second-moment formula of Harlow, Usatyuk, and Zhao therefore yields

$$\mathbb{E}_O \left| \langle \phi | \widehat{V}^\dagger \widehat{V} | \psi \rangle \right|^2 = \frac{D}{D+2} \text{Tr} \rho_P^2, \quad (10)$$

so equation (2) becomes a field-energy lower floor on this Haar-averaged code fluctuation. This is an ensemble mean-square statement for the specified pair, not a deterministic bound for every fixed code.

Finally, we give a limited gravitational application. If every conditional spherical $q_i = 0$ branch obeys the final-slice local constraint budget $Q_{b,i} \leq \delta$, then

$$S_2(\rho_P) \leq \delta \frac{R^2}{G} \frac{C_{\text{opt}}(y) \tanh y \text{sech}^2 y}{2}. \quad (11)$$

The budget must hold branchwise: controlling only an averaged or centered source does not control the conditional geometries. This result composes a fixed-background channel calculation with exact Einstein–scalar final-slice constraints. It neither recomputes the channel on the branch-dependent geometries nor solves a coupled evolution.

Compact final support is essential throughout. Without it, arbitrarily soft thermal modes prevent the energy norm from controlling the dephasing form. If a source is confined to areal radius $a < R$ for a static-time interval of length T , finite propagation supplies the causal envelope

$$L = R \arctanh(a/R) + T. \quad (12)$$

Smooth compact sources approach the final-support optimum when their worldtube approaches the final-support ball. We do not claim that every fixed smaller source cylinder can approach the top mode.

The field resource in this paper is equally specific. It is the post-switch scalar Killing energy of the prescribed conditional sources. It does not include a clock, trap, battery, actuator, or autonomous observer stress. Exact gapless-detector channels and their energy accounting are prior art [2, 3, 4]; positive compact kernels and logarithmic-potential variational principles also have a substantial literature. The candidate contribution is the finite-pointer purity theorem with the exact localized thermal coefficient, its all-sector de Sitter reduction, and the explicit insertion of that physical purity into the observer-code second moment. External detector/QFT, operator-theory, and quantum-gravity review remain publication gates.

Section 2 defines the controlled-source model and exact Schur channel. Section 3 proves the localized thermal operator theorem and the full de Sitter reduction. Section 4 derives the Rényi bound and observer-code insertion. Section 5 treats smooth source realization, section 6 separates the result from adjacent entropy and detector bounds, and section 7 states the claim boundary. Appendix A contains the branchwise final-slice application.

2 Static-patch model and exact pointer channel

2.1 Optical partial waves

In static coordinates, four-dimensional de Sitter space has

$$ds^2 = -N(r) dt^2 + \frac{dr^2}{N(r)} + r^2 d\Omega_2^2, \quad N(r) = 1 - \frac{r^2}{R^2}, \quad 0 \leq r < R. \quad (13)$$

Introduce the optical radius

$$x = R \arctanh(r/R), \quad r = R \tanh(x/R), \quad \Omega(x) = \text{sech}(x/R). \quad (14)$$

Then $g_{\text{dS}} = \Omega^2 g_{\text{opt}}$, where

$$g_{\text{opt}} = -dt^2 + dx^2 + R^2 \sinh^2(x/R) d\Omega_2^2. \quad (15)$$

For the conformally coupled massless scalar, spherical-harmonic decomposition and the standard radial rescaling give the positive operators

$$A_\ell = -\frac{d^2}{dx^2} + \frac{\ell(\ell+1)}{R^2 \sinh^2(x/R)}, \quad h_\ell = A_\ell^{1/2} \quad (16)$$

on $L^2(\mathbb{R}_+)$, with the regular Dirichlet condition at $x = 0$. The Bunch–Davies state restricted to the static patch is KMS at

$$\beta = 2\pi R \quad (17)$$

for the static Killing flow. The half-line theorem in section 3 first keeps β arbitrary; theorem 3.5 then imposes this de Sitter relation when comparing all angular and canonical sectors.

Let $(q_{\ell m}, p_{\ell m})$ denote real final Cauchy data for the rescaled partial waves. For smooth compact data, the thermal quadratic form and the Killing energy are

$$\Gamma_{\ell m} = \langle q_{\ell m}, h_\ell \coth(\beta h_\ell/2) q_{\ell m} \rangle + \langle p_{\ell m}, h_\ell^{-1} \coth(\beta h_\ell/2) p_{\ell m} \rangle, \quad (18)$$

$$E_{\ell m} = \frac{1}{2} \left(\|h_\ell q_{\ell m}\|^2 + \|p_{\ell m}\|^2 \right). \quad (19)$$

We write $\Gamma = \sum_{\ell m} \Gamma_{\ell m}$ and $E_K = \sum_{\ell m} E_{\ell m}$. Smooth angular data make these sums finite in the needed form domains.

For $\ell = 0$, the unitary sine transform diagonalizes h_0 . With hats denoting that transform,

$$\Gamma_{00} = \int_0^\infty \coth(\beta k/2) \left(k |\widehat{q}(k)|^2 + \frac{|\widehat{p}(k)|^2}{k} \right) dk, \quad (20)$$

$$E_{00} = \frac{1}{2} \int_0^\infty \left(k^2 |\widehat{q}(k)|^2 + |\widehat{p}(k)|^2 \right) dk. \quad (21)$$

2.2 Local interaction and exact channel

Let P be a finite pointer with orthonormal basis $\{|i\rangle\}_{i=1}^d$. Assign a real compact source J_i to each pointer state and couple it to the physical scalar by

$$S_{\text{int}} = - \int_{\mathcal{M}} \sqrt{-g} \phi(X) \sum_{i=1}^d J_i(X) |i\rangle\langle i| d^4X, \quad J_i \in C_c^\infty(\mathcal{M}; \mathbb{R}). \quad (22)$$

The switching functions and spatial smearings are included in the J_i . The pointer projectors commute, and free-field commutators are c-numbers, so the Magnus series terminates in every controlled sector. Up to source-dependent phases,

$$U = \sum_{i=1}^d e^{i\Xi[J_i]} |i\rangle\langle i| \otimes e^{-i\phi(J_i)}. \quad (23)$$

This standard exact evolution is reviewed in [2, 3, 4]. The interaction is compactly supported, and its densities commute at spacelike separation because the diagonal pointer operators commute and the scalar field is microcausal. Nevertheless, the same finite pointer appears across each spatial smearing. Generic spatially extended nonrelativistic detectors can violate causal factorization [5]; fully relativistic localized

probe fields provide a systematic replacement framework [6]. We treat equation (22) as a prescribed gapless-detector idealization, not as an autonomous pointer field or material actuator.

Let E denote the causal propagator and K the positive-frequency map for the static one-particle structure. Purify the Bunch–Davies KMS state and let $|e_i\rangle$ be the conditional purified field state. Their Gram matrix $G_{ij} = \langle e_j, e_i \rangle$ is positive semidefinite with unit diagonal. Tracing the field gives the exact Schur dephasing channel

$$\mathcal{D}_G(|i\rangle\langle j|) = G_{ij} |i\rangle\langle j|, \quad |G_{ij}| = e^{-\Gamma_{ij}}. \quad (24)$$

Let (q_i, p_i) be the complete conditional final data, and write $q_{ij} = q_i - q_j$, $p_{ij} = p_i - p_j$. Gaussianity gives

$$\Gamma_{ij} = \frac{1}{4} \sum_{\ell m} [\langle q_{ij, \ell m}, h_\ell \coth(\beta h_\ell/2) q_{ij, \ell m} \rangle + \langle p_{ij, \ell m}, h_\ell^{-1} \coth(\beta h_\ell/2) p_{ij, \ell m} \rangle]. \quad (25)$$

For an initial pointer state $|\psi\rangle = \sum_i \sqrt{w_i} |i\rangle$, the reduced pointer state is

$$\rho_{P, ij} = \sqrt{w_i w_j} G_{ij}, \quad \text{Tr } \rho_P^2 = \sum_{i, j} w_i w_j e^{-2\Gamma_{ij}}. \quad (26)$$

The binary model is recovered by $d = 2$, $J_\pm = \pm J$. Its conditional data are opposite, so equation (25) gives $\Gamma_{+-} = \Gamma$ with

$$\Gamma = 2 \langle KEJ, \coth(\beta h/2) KEJ \rangle. \quad (27)$$

The phase of $G_{+-} = \kappa$ is a correctable pointer rotation. If $\mathcal{D}_0$ denotes complete binary dephasing, then

$$\epsilon_{\text{obs}} := \frac{1}{2} \|\mathcal{D}_\kappa - \mathcal{D}_0\|_\diamond = \frac{1}{2} |\kappa| = \frac{1}{2} e^{-\Gamma}. \quad (28)$$

The equality follows from $\mathcal{D}_\kappa - \mathcal{D}_0 = (\kappa/2)(\text{id} - \text{Ad}_{Z_P})$ after the correctable phase is removed, and it is saturated by a pointer superposition. The exact quantum-to-classical channel used in the observer rule of Harlow, Usatyuk, and Zhao [1] is the limit in which all off-diagonal G_{ij} vanish. Rather than identify the binary ϵ_{obs} with a gravitational encoding error, section 4 below uses the physical purity in equation (26), which is the quantity that enters their second-Rényi code calculation.

After switching, branch i has excess scalar Killing energy E_i and renormalized-stress excess

$$E_i = \frac{1}{2} \sum_{\ell m} (\|h_\ell q_{i, \ell m}\|^2 + \|p_{i, \ell m}\|^2), \quad \Delta \langle T_{ab}^{\text{ren}} \rangle_i = T_{ab}[\phi_{J_i}]. \quad (29)$$

Let $\bar{q} = \sum_i w_i q_i$, $\bar{p} = \sum_i w_i p_i$. The energy that controls pairwise separation is

$$\bar{E}_K := \sum_i w_i E[q_i - \bar{q}, p_i - \bar{p}] = \sum_i w_i E_i - E[\bar{q}, \bar{p}]. \quad (30)$$

For the symmetric binary model, $\bar{q} = \bar{p} = 0$ and $E_+ = E_- = \bar{E}_K = E_K$. The state-independent Hadamard subtraction cancels in the difference, and the background one-point function vanishes. These quantities price the post-switch scalar fields generated by the same J_i as the channel. They do not include the stress of a clock, trap, or material device that realizes the prescribed coupling.

2.3 Causal support

Suppose every J_i is supported inside the centered areal ball $r \leq a < R$ during a common static-time interval of length T . Its optical radius is

$$\ell = R \operatorname{arctanh}(a/R). \quad (31)$$

Finite propagation in the optical metric implies that the final Cauchy data of every partial wave are supported in

$$0 \leq x \leq L, \quad L = \ell + T. \quad (32)$$

No spectral cutoff or hard spatial wall is introduced.

3 Sharp final-support thermal bounds

Let P_L be multiplication by $(0, L)$ on $L^2(\mathbb{R}_+)$. We first solve the momentum problem at arbitrary inverse temperature β . We then impose the de Sitter relation $\beta = 2\pi R$ and prove that the same momentum sector is the optimum over every angular and canonical sector.

Lemma 3.1 (Compressed half-line inverses). *For $h_0 = \sqrt{-\partial_x^2}$ with the half-line Dirichlet condition,*

$$(P_L h_0^{-1} P_L)(x, y) = \frac{1}{\pi} \log \frac{x+y}{|x-y|}, \quad (33)$$

$$(P_L h_0^{-2} P_L)(x, y) = \min(x, y). \quad (34)$$

Moreover,

$$\|P_L h_0^{-1} P_L\| \leq \frac{2 \operatorname{arsinh}(1)}{\pi} L, \quad (35)$$

$$\|P_L h_0^{-2} P_L\| = \frac{4}{\pi^2} L^2. \quad (36)$$

Proof. The unitary sine transform gives

$$\frac{2}{\pi} \int_0^\infty \frac{\sin(kx) \sin(ky)}{k} dk = \frac{1}{\pi} \log \frac{x+y}{|x-y|}$$

and

$$\frac{2}{\pi} \int_0^\infty \frac{\sin(kx) \sin(ky)}{k^2} dk = \min(x, y).$$

The logarithmic kernel is positive. With $u = x/L$, its row integral divided by L is

$$\frac{1}{\pi} [(1+u) \log(1+u) - 2u \log u - (1-u) \log(1-u)]. \quad (37)$$

Its maximum is $2 \operatorname{arsinh}(1)/\pi$ at $u = 1/\sqrt{2}$, so Schur's test proves equation (35). The kernel $\min(x, y)$ is the inverse of $-\partial_x^2$ on $(0, L)$ with Dirichlet condition at zero and Neumann condition at L . Its first eigenvalue is $(\pi/2L)^2$, proving equation (36). $\square$

Lemma 3.2 (Exact localized KMS kernel). *Set*

$$B_{\beta, L} := P_L h_0^{-1} \coth(\beta h_0/2) P_L. \quad (38)$$

Its integral kernel is

$$B_{\beta, L}(x, y) = \frac{1}{\pi} \log \frac{\sinh[\pi(x+y)/\beta]}{\sinh[\pi|x-y|/\beta]}. \quad (39)$$

Let $\tau = \pi L/\beta$ and define on $L^2(0, 1)$

$$k_\tau(u, v) := \frac{1}{\pi} \log \frac{\sinh[\tau(u+v)]}{\sinh[\tau|u-v|]}, \quad \Lambda(\tau) := \|\mathcal{K}_\tau\|. \quad (40)$$

Then $B_{\beta, L}$ is unitarily equivalent to $L\mathcal{K}_\tau$. $\mathcal{K}_\tau$ is positive, compact, and positivity improving. Its top eigenvalue $\Lambda(\tau)$ is simple and has a strictly positive eigenfunction.

Proof. Using $\coth z = 1 + 2 \sum_{n \geq 1} e^{-2nz}$ and integrating each sine-transform term gives

$$\begin{aligned} \frac{2}{\pi} \int_0^\infty \frac{\sin(kx) \sin(ky)}{k} \coth(\beta k/2) dk &= \frac{1}{\pi} \log \frac{x+y}{|x-y|} \\ &+ \frac{1}{\pi} \sum_{n \geq 1} \log \frac{(n\beta)^2 + (x+y)^2}{(n\beta)^2 + (x-y)^2}. \end{aligned} \quad (41)$$

Euler's product for $\sinh$ reduces this expression to equation (39). Scaling $x = Lu$, $y = Lv$ gives the unitary equivalence. The logarithmic diagonal singularity is square integrable on the unit square, so the operator is Hilbert–Schmidt. The spectral definition makes it positive, while the displayed kernel is strictly positive away from the measure-zero boundary and diagonal. The compact positivity-improving operator therefore has a simple positive top eigenfunction by the Perron–Frobenius–Jentzsch theorem. $\square$

Theorem 3.3 (Sharp thermal half-line momentum bound). *Let $\beta, L > 0$, let $h_0 = \sqrt{-\partial_x^2}$ on the Dirichlet half-line, and define*

$$C_\beta(L) := \sup_{\substack{p \neq 0 \\ \text{supp } p \subset [0, L]}} \frac{\langle p, h_0^{-1} \coth(\beta h_0/2) p \rangle}{\|p\|^2 / 2}. \quad (42)$$

With $\tau = \pi L/\beta$,

$$\boxed{C_\beta(L) = 2L\Lambda(\tau)}. \quad (43)$$

The normalized optimizer in the finite-energy closure is unique up to sign and is the positive top eigenfunction of $\mathcal{K}_\tau$. The global bounds

$$\max \left\{ \frac{3L}{\pi}, \frac{16L^2}{\beta\pi^2} \right\} \leq C_\beta(L) \leq \frac{4 \operatorname{arsinh}(1)}{\pi} L + \frac{16L^2}{\beta\pi^2} \quad (44)$$

hold for every $\beta, L > 0$. If

$$M(\tau) := \max_{0 < u < 1} \int_0^1 k_\tau(u, v) dv, \quad (45)$$

then $C_\beta(L) \leq 2LM(\tau)$, and the maximizing row is at

$$u_\tau = \frac{1}{\tau} \operatorname{arsinh} \left(\frac{\sinh \tau}{\sqrt{2}} \right). \quad (46)$$

Finally,

$$0 \leq C_\beta(L) - 2L\Lambda(0) \leq \frac{2\pi L^3}{3\beta^2}, \quad (47)$$

$$0 \leq C_\beta(L) - \frac{16L^2}{\beta\pi^2} \leq \frac{\beta}{3}. \quad (48)$$

Thus the exact answer depends on temperature and support only through $\tau = \pi L/\beta$, apart from its overall length scale.

Proof. By lemma 3.2, the quotient in equation (42) has supremum $2\|B_{\beta, L}\| = 2L\Lambda(\tau)$, with the stated unique optimizer. The profile $p(x) = \sqrt{3/L^3} x$ gives the vacuum lower bound $3L/\pi$, while the first eigenfunction of the Dirichlet–Neumann Green operator and $\coth z \geq z^{-1}$ give $16L^2/(\beta\pi^2)$. Conversely,

$\coth z \leq 1 + z^{-1}$ and lemma 3.1 give the upper bound in equation (44). Positivity and Schur's test give the $M(\tau)$ bound. Differentiating the row integral and using

$$\sinh[\tau(1+u)] \sinh[\tau(1-u)] = \sinh^2 \tau - \sinh^2(\tau u) \quad (49)$$

gives its unique stationary point equation (46).

For small τ , write $k_\tau = k_0 + d_\tau$. Euler's product implies, for $A \geq B \geq 0$,

$$0 \leq \log \frac{\sinh A/A}{\sinh B/B} \leq \frac{A^2 - B^2}{6}. \quad (50)$$

Taking $A = \tau(u+v)$ and $B = \tau|u-v|$ and applying Schur's test gives $0 \leq \Lambda(\tau) - \Lambda(0) \leq \tau^2/(3\pi)$, which proves equation (47). For large τ ,

$$k_\tau(u, v) = \frac{2\tau}{\pi} \min(u, v) + r_\tau(u, v). \quad (51)$$

The remainder has the explicit expansion

$$r_\tau(u, v) = \frac{1}{\pi} \sum_{n \geq 1} \frac{1}{n} \left[e^{-2n\tau|u-v|} - e^{-2n\tau(u+v)} \right]. \quad (52)$$

Each bracket is a positive multiple of the Dirichlet half-line resolvent kernel of $-\partial_u^2 + (2n\tau)^2$, so $r_\tau \geq 0$ in operator order. For $a > 0$,

$$\int_0^\infty \left[e^{-a|u-v|} - e^{-a(u+v)} \right] dv = \frac{2(1 - e^{-au})}{a} \leq \frac{2}{a}. \quad (53)$$

Monotone summation and Schur's test therefore give $\|r_\tau\| \leq (\pi\tau)^{-1} \sum_{n \geq 1} n^{-2} = \pi/(6\tau)$. Since the norm of the $\min(u, v)$ operator is $4/\pi^2$, multiplication by $2L$ proves equation (48). $\square$

The angular potential improves the complete thermal momentum form, not only the two estimates in lemma 3.1.

Lemma 3.4 (All-angular and coordinate domination). *For every $\ell \geq 0$, $A_\ell \geq A_0$. Moreover,*

$$\begin{aligned} h_\ell^{-1} \coth(\beta h_\ell/2) &= \frac{2}{\beta} A_\ell^{-1} + \frac{4}{\beta} \sum_{n \geq 1} [A_\ell + (2\pi n/\beta)^2]^{-1} \\ &\leq h_0^{-1} \coth(\beta h_0/2) \end{aligned} \quad (54)$$

as generalized quadratic forms. If q is supported in $[0, L]$, then

$$\|q\| \leq \frac{L}{\pi} \|h_\ell q\|, \quad (55)$$

$$\langle q, h_\ell q \rangle \leq \frac{L}{\pi} \|h_\ell q\|^2, \quad (56)$$

$$\frac{\Gamma_{q, \ell m}}{E_{q, \ell m} R} \leq \frac{2y}{\pi} + \frac{2y^2}{\pi^3}, \quad y = L/R, \quad \beta = 2\pi R. \quad (57)$$

Proof. The nonnegative angular potential gives $A_\ell \geq A_0$. The Mittag–Leffler expansion $\coth z/z = z^{-2} + 2 \sum_{n \geq 1} (z^2 + \pi^2 n^2)^{-1}$ gives equation (54); every resolvent is order decreasing. Extending q by zero places it in $H_0^1(0, L)$. Dirichlet Poincare and $\|h_\ell q\|^2 = \langle q, A_\ell q \rangle$ give equation (55); Cauchy–Schwarz gives equation (56). Finally, $\coth z \leq 1 + z^{-1}$ yields

$$\Gamma_{q, \ell m} \leq \left[\frac{L}{\pi} + \frac{2L^2}{\beta \pi^2} \right] \|h_\ell q\|^2. \quad (58)$$

Divide by $E_{q, \ell m} = \|h_\ell q\|^2 / 2$ and set $\beta = 2\pi R$. No locality of the nonlocal vector $h_\ell q$ is assumed. $\square$

We can now specialize the half-line theorem to the static patch and prove that no angular or field-coordinate sector gives a larger quotient. The supremum below is unchanged if smooth compact data are replaced by their finite-energy closure.

Theorem 3.5 (Conformal de Sitter all-sector optimum). *Let*

$$C_{\text{opt}}(y) := \sup \left\{ \frac{\Gamma}{E_K R} : 0 < E_K < \infty, \text{supp}(q, p) \subset [0, yR] \right\}. \quad (59)$$

For the full conformally coupled massless scalar in the Bunch–Davies state, including arbitrary angular data,

$$\boxed{C_{\text{opt}}(y) = 2y\Lambda(y/2)}. \quad (60)$$

Equivalently,

$$RC_{\text{opt}}(L/R) = C_{2\pi R}(L). \quad (61)$$

The supremum lies in the s-wave momentum sector. Its normalized optimizer is the unique positive top eigenfunction of equation (40), up to an overall sign. Specializing equation (44) gives

$$\max \left\{ \frac{3y}{\pi}, \frac{8y^2}{\pi^3} \right\} \leq C_{\text{opt}}(y) \leq F(y), \quad (62)$$

$$F(y) = \frac{4 \operatorname{arsinh}(1)}{\pi} y + \frac{8}{\pi^3} y^2. \quad (63)$$

The sharper bound $C_{\text{opt}}(y) \leq 2yM(y/2)$ also follows from theorem 3.3. Its global remainders become

$$0 \leq C_{\text{opt}}(y) - 2y\Lambda(0) \leq \frac{y^3}{6\pi}, \quad (64)$$

$$0 \leq C_{\text{opt}}(y) - \frac{8y^2}{\pi^3} \leq \frac{2\pi}{3}. \quad (65)$$

Thus temperature first corrects the localized vacuum cost at cubic order, while $8y^2/\pi^3$ is the sharp leading large-support coefficient.

Proof. For momentum data in the s-wave, theorem 3.3 gives

$$\sup_{p \neq 0} \frac{\Gamma_p}{E_p R} = \frac{C_{2\pi R}(L)}{R} = 2y\Lambda(y/2). \quad (66)$$

Equation (54) shows that no higher angular momentum sector is worse. The linear trial profile in equation (86) gives the lower bound $3y/\pi$. Also $\coth z \geq z^{-1}$ and equation (36) give

$$\sup_{p \neq 0} \frac{\Gamma_p}{E_p R} \geq \frac{8y^2}{\pi^3}. \quad (67)$$

The right side of equation (57) is bounded by $3y/\pi$ when $y \leq \pi^2/2$, and by $8y^2/\pi^3$ when $y \geq \pi^2/3$. These ranges overlap, so the entire coordinate sector is dominated by the s-wave momentum supremum for every $y > 0$. The domination is strict for nonzero coordinate data. Indeed, at finite β , the thermal momentum form is strictly larger than its vacuum part and strictly larger than $(2/\beta)A_0^{-1}$ on every nonzero vector. Applying those two strict inequalities to the trial profiles used above covers the same two overlapping ranges.

The angular domination is also strict for $\ell > 0$. For any $c > 0$, the resolvent difference $(A_0+c)^{-1} - (A_\ell+c)^{-1}$ has a strictly positive quadratic form on nonzero data: equality would force $V_\ell^{1/2}(A_0+c)^{-1}p = 0$, where $V_\ell = \ell(\ell+1)/[R^2 \sinh^2(x/R)] > 0$ almost everywhere, and hence $p = 0$. The $n = 1$ term of equation (54) therefore excludes a higher-angular-momentum tie. Together with the simple top eigenvalue of $\mathcal{K}_{y/2}$, this proves the asserted uniqueness as well as equation (60). The explicit bracket and both remainder bounds are equations (44) and (47) and equation (48) with $\beta = 2\pi R$ and $L = yR$. $\square$

3.1 Numerical realization of the sharp coefficient

The compact-operator formula can be evaluated directly without entering the proof. We use normalized piecewise constants on a uniform partition of $(0, 1)$. Cell integrals of the vacuum kernel are evaluated exactly from

$$\frac{d^2}{dz^2} \left(\frac{z^2}{2} \log |z| - \frac{3z^2}{4} \right) = \log |z|, \quad (68)$$

while Gauss–Legendre quadrature is applied only to the smooth thermal correction $\log[\sinh z/z]$. Figure 1 shows the resulting 128-cell Rayleigh–Ritz curve and positive top eigenfunctions.

At $y = 0.1, 1, 10$ the computed costs are respectively 0.0997264, 1.02960, and 27.7170. Across those three points, the largest relative change from 64 to 128 cells is 5.0×10^{-5} ; changing the thermal quadrature order from 8 to 12 at 64 cells changes the values by at most 1.1×10^{-15} . Every one of the nineteen plotted estimates lies inside equation (62) and below the exact-kernel Schur bound. Because the thermal quadrature is not interval enclosed, the plotted curve is numerical evidence rather than an additional rigorous lower bound; none of the theorem statements depends on it.

Combining equations (28), (32) and (60) gives the operational form.

Corollary 3.6 (Causal source-cylinder envelope). *If the source in equation (22) is supported in $r \leq a < R$ for a static-time interval of length T , set*

$$y = \operatorname{arctanh}(a/R) + T/R. \quad (69)$$

Then

$$\log \frac{1}{2\epsilon_{\text{obs}}} \leq E_K R C_{\text{opt}}(y) = 2E_K R y \Lambda(y/2) \leq E_K R F(y). \quad (70)$$

The first coefficient is sharp for the relaxed problem in which only the final Cauchy support L is fixed. For a cylinder with smaller spatial radius $\ell = L - T$, the bound remains valid, but reachability of its top eigenfunction is not asserted. Exact complete dephasing at finite a , T , and R requires $E_K = \infty$.

Remark 3.7. The final statement is a no-go only for exact complete dephasing in the frozen local scalar model. It is not a universal energy cost for quantum measurement, and it does not price the external device used to realize J .

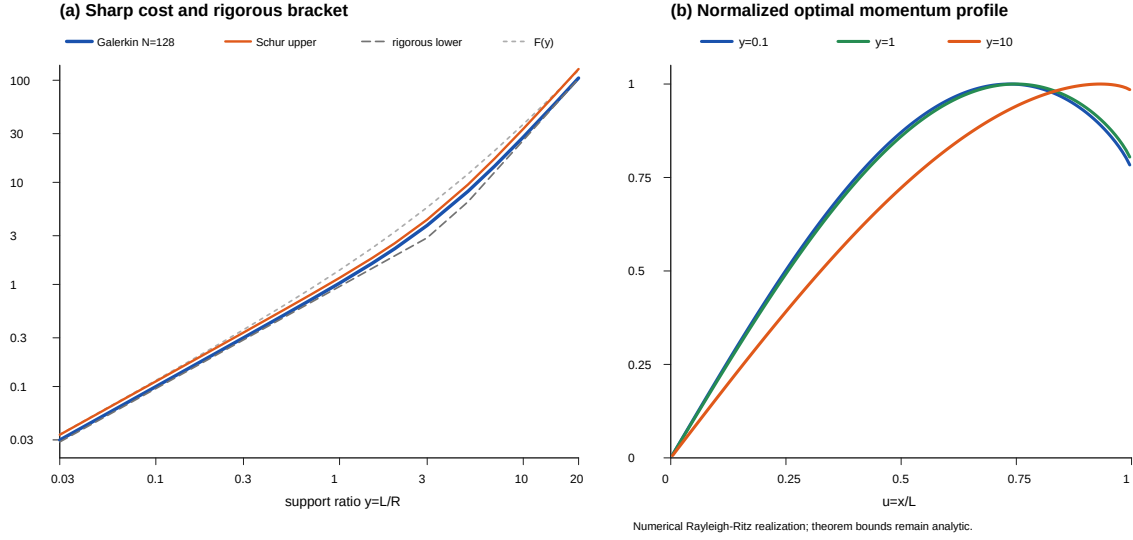


Figure 1: Numerical realization of the exact compact-kernel theorem. Panel (a) compares the 128-cell Galerkin estimate with the rigorous analytic lower bound, the exact-kernel row-Schur upper bound, and the simpler closed-form upper bound $F(y)$. Panel (b) shows the positive top eigenfunction, normalized by its maximum, at three support ratios.

4 Finite-pointer entropy and observer-code accuracy

We now compose the sharp binary covariance theorem with the exact finite-pointer channel. The composition uses no new spectral estimate. Its input is the pairwise form bound; its additional ingredients are the weighted variance identity and convexity.

Define the ideal classical pointer purity and its second Rényi entropy by

$$P_{\text{cl}} := \sum_i w_i^2, \quad H_2(w) := -\log P_{\text{cl}}. \quad (71)$$

Theorem 4.1 (Finite-pointer energy–support–entropy bound). *Let p_i be Dirichlet half-line momentum data supported in $[0, L]$, with $w_i \geq 0$ and $\sum_i w_i = 1$. Set*

$$\bar{p} = \sum_i w_i p_i, \quad \bar{E}_K = \frac{1}{2} \sum_i w_i \|p_i - \bar{p}\|^2. \quad (72)$$

For the exact thermal channel in equation (24),

$$\boxed{\text{Tr } \rho_P^2 \geq P_{\text{cl}} + (1 - P_{\text{cl}}) \exp \left[-\frac{C_\beta(L) \bar{E}_K}{1 - P_{\text{cl}}} \right]} \quad (73)$$

when $P_{\text{cl}} < 1$, with the continuous value one when $P_{\text{cl}} = 1$. Consequently,

$$S_2(\rho_P) \leq -\log \left\{ P_{\text{cl}} + (1 - P_{\text{cl}}) \exp \left[-\frac{C_\beta(L) \bar{E}_K}{1 - P_{\text{cl}}} \right] \right\} \quad (74)$$

$$\leq \min\{H_2(w), C_\beta(L) \bar{E}_K\}. \quad (75)$$

The first inequality is saturated for equal binary weights and opposite top-eigenfunction profiles. Global sharpness for arbitrary pointer dimension is not asserted.

Proof. Write $d_{ij} = p_i - p_j$ and $x_{ij} = 2\Gamma_{ij}$. By equations (25) and (42),

$$x_{ij} = \frac{1}{2} \langle d_{ij}, B_{\beta,L} d_{ij} \rangle \leq \frac{C_{\beta}(L)}{4} \|d_{ij}\|^2. \quad (76)$$

The weighted variance identity gives

$$\sum_{i,j} w_i w_j \|p_i - p_j\|^2 = 4\bar{E}_K, \quad (77)$$

and hence

$$\sum_{i,j} w_i w_j x_{ij} \leq C_{\beta}(L) \bar{E}_K. \quad (78)$$

The terms $i = j$ have $x_{ii} = 0$ and total weight P_{cl} . Normalize the remaining weights by $1 - P_{\text{cl}}$ and apply Jensen's inequality to e^{-x} in equation (26); this proves equation (73). The first simple bound follows because the right side is at least P_{cl} . For $z \geq 0$, convexity also gives

$$P_{\text{cl}} + (1 - P_{\text{cl}})e^{-z/(1-P_{\text{cl}})} \geq e^{-z}. \quad (79)$$

Taking $z = C_{\beta}(L)\bar{E}_K$ proves the second bound in equation (75).

For two equal weights and $p_{\pm} = \pm p$, the off-diagonal exponents coincide. If p is the top eigenfunction of $B_{\beta,L}$, both the form bound and Jensen's inequality are equalities. For more pointer states, equality would require all nonzero differences to lie in the simple top eigenspace and to have equal exponent. We do not claim that those conditions can be met. $\square$

The same argument works in the full static-patch phase space because the energy is a Hilbert norm and theorem 3.5 bounds the thermal form of every difference datum.

Corollary 4.2 (Conformal de Sitter finite-pointer bound). *Let every conditional angular and canonical datum be supported in $[0, yR]$, and define $\bar{E}_K$ by centering in the full energy norm as in equation (30). Then*

$$\text{Tr } \rho_P^2 \geq P_{\text{cl}} + (1 - P_{\text{cl}}) \exp \left[-\frac{RC_{\text{opt}}(y)\bar{E}_K}{1 - P_{\text{cl}}} \right], \quad (80)$$

$$S_2(\rho_P) \leq \min\{H_2(w), RC_{\text{opt}}(y)\bar{E}_K\}. \quad (81)$$

The coefficient $C_{\text{opt}}(y) = 2y\Lambda(y/2)$ is exact. Binary saturation uses the unique s -wave momentum optimizer of theorem 3.5.

4.1 Insertion into the observer code

The field record provides a purification of the physical pointer state:

$$|\omega\rangle = \sum_i \sqrt{w_i} |i\rangle_{\text{Ob}} |e_i\rangle_{\text{Ob}'}, \quad \langle e_j, e_i \rangle = G_{ij}. \quad (82)$$

The two reductions of this pure state have identical nonzero spectra, so

$$e^{-S_2(\omega_{\text{Ob}'})} = \text{Tr } \omega_{\text{Ob}'}^2 = \text{Tr } \rho_P^2. \quad (83)$$

This is the entropy parameter appearing in the simple random-code calculation of Harlow, Usatyuk, and Zhao [1].

Proposition 4.3 (Physical floor in the simple observer code). *Let D be the dimension of their Haar-random orthogonal encoding matrix, and let $|\phi\rangle, |\psi\rangle$ be orthogonal CRT-real matter states. Replacing the ideal record by equation (82), their exact second-moment formula gives*

$$\mathbb{E}_O \left| \langle \phi | \widehat{V}^\dagger \widehat{V} | \psi \rangle \right|^2 = \frac{D}{D+2} \text{Tr} \rho_P^2. \quad (84)$$

Consequently, for the de Sitter model,

$$\mathbb{E}_O \left| \langle \phi | \widehat{V}^\dagger \widehat{V} | \psi \rangle \right|^2 \geq \frac{D}{D+2} \left\{ P_{\text{cl}} + (1 - P_{\text{cl}}) e^{-RC_{\text{opt}}(y) \bar{E}_K / (1 - P_{\text{cl}})} \right\}. \quad (85)$$

Proof. Equation (4.2) of [1] contains the observer terms $e^{-S_2(\omega_{\text{ob}'})}$ and $\text{Tr}(\omega_{\text{ob}} \omega_{\text{ob}}^T)$ together with two matter overlaps. For the stated orthogonal CRT-real pair, both overlaps vanish, leaving equation (84). Apply equations (80) and (83). $\square$

The proposition closes a specific gap: the entropy suppressing the simple code fluctuation is now generated by a localized field record with an exact energy–support bound. Its scope is narrower than a gravitational observer theorem. The system called Ob' in equation (82) is the purified nonideal record, and equation (85) is a Haar-ensemble mean-square statement for one orthogonal pair. It is not a deterministic lower bound for every fixed encoding map. Nor does it include an autonomous observer, actuator work, or gravitational evolution.

5 Optimal profiles and smooth source realization

5.1 Explicit trial profiles

The exact optimizer in theorem 3.5 is the top eigenfunction of $\mathcal{K}_{y/2}$. Two elementary profiles give the explicit lower bounds in equation (62) and make both asymptotic regimes concrete. For the vacuum-controlled regime, set $L = yR$, choose $q = 0$, and use

$$p_L(x) = \sqrt{\frac{3}{L^3}} x \mathbf{1}_{(0,L)}(x). \quad (86)$$

The vacuum part of the thermal form is

$$\begin{aligned} \langle p_L, h_0^{-1} p_L \rangle &= \frac{3}{\pi L^3} \int_0^L \int_0^L xy \log \frac{x+y}{|x-y|} dx dy \\ &= \frac{3L}{2\pi}. \end{aligned} \quad (87)$$

For the second equality, scale $x = Lu$, $y = Lv$, split the unit square across the diagonal, and use

$$\int_0^1 t \log \frac{1+t}{1-t} dt = 1. \quad (88)$$

Since $\coth(\beta h_0/2) \geq 1$, the full thermal dephasing exponent is at least the value in equation (87). On the other hand, $E_K = 1/2$ for the normalized datum. Thus $\Gamma/(E_K R) \geq 3L/(\pi R) = 3y/\pi$.

For the thermal regime, use instead the normalized first eigenfunction of the Dirichlet–Neumann Green operator,

$$p_L^{\text{th}}(x) = \sqrt{\frac{2}{L}} \sin \frac{\pi x}{2L} \mathbf{1}_{(0,L)}(x). \quad (89)$$

Since $\coth z \geq z^{-1}$ and $\langle p_L^{\text{th}}, h_0^{-2} p_L^{\text{th}} \rangle = 4L^2/\pi^2$, this datum gives

$$\frac{\Gamma}{E_K R} \geq \frac{8y^2}{\pi^3}. \quad (90)$$

The first profile and the vacuum Schur bound also show

$$\frac{3}{\pi} \leq \liminf_{y \downarrow 0} \frac{C_{\text{opt}}(y)}{y} \leq \limsup_{y \downarrow 0} \frac{C_{\text{opt}}(y)}{y} \leq \frac{4 \operatorname{arsinh}(1)}{\pi}. \quad (91)$$

The ratio of the analytic upper and lower linear coefficients is $4 \operatorname{arsinh}(1)/3 \simeq 1.17516$.

For a target exponent $\Gamma_\star$, scaling equation (86) by an amplitude A yields the sufficient energy bound

$$E_K \leq \frac{\pi \Gamma_\star}{3L}. \quad (92)$$

This is an upper bound on the energy needed by this simple family. The sharp half-line momentum infimum at inverse temperature β is instead $\Gamma_\star/C_\beta(L) = \Gamma_\star/[2L\Lambda(\pi L/\beta)]$. At the de Sitter temperature this is $\Gamma_\star/[RC_{\text{opt}}(L/R)]$.

5.2 Smooth compact sources

The hard endpoint in equation (86) is useful for exact constants but is not a smooth source. We now show that it is a legitimate limiting profile and that every strict window persists.

Lemma 5.1 (Smooth density closure). *Fix an optical source radius $\ell > 0$ and a normalized $p \in L^2(0, \ell)$. There are normalized $p_n \in C_c^\infty([0, \ell])$, regular at the origin, such that*

$$p_n \longrightarrow p \quad \text{in } L^2(\mathbb{R}_+). \quad (93)$$

Their thermal dephasing forms converge, and their cumulative energy measures converge uniformly.

Proof. Density of smooth functions supported strictly inside $(0, \ell)$ gives equation (93); such functions are automatically regular radial data at the origin. On data supported in $[0, \ell]$,

$$P_\ell h_0^{-1} \coth(\beta h_0/2) P_\ell \leq P_\ell h_0^{-1} P_\ell + \frac{2}{\beta} P_\ell h_0^{-2} P_\ell \quad (94)$$

as quadratic forms. The right side is bounded by lemma 3.1; hence the covariance is continuous under L^2 convergence. Finally,

$$\sup_{0 \leq x \leq \ell} \left| \int_0^x (p_n^2 - p^2) \, ds \right| \leq \|p_n - p\|_2 (\|p_n\|_2 + \|p\|_2), \quad (95)$$

which proves uniform convergence of cumulative energy. $\square$

It remains to produce a supported final datum by a smooth spacetime source. This also proves sharpness over sources when the source ball approaches the declared final-support ball; it does not prove exact controllability from a strictly smaller cylinder of optical radius $L - T$. Let $\phi_{\text{free},n}$ be the homogeneous solution with final data $(q, p) = (0, p_n)$. Choose a smooth time cutoff η_n that vanishes before the protocol, equals one near the final slice, and changes over a time shorter than the optical gap between $\text{supp } p_n$ and ℓ . If $\mathcal{P}$ is the conformal wave operator, set

$$J_n = \mathcal{P}(\eta_n \phi_{\text{free},n}). \quad (96)$$

The retarded solution sourced by J_n is $\eta_n \phi_{\text{free},n}$. It therefore has exactly the desired final data. Finite propagation keeps J_n inside the declared optical worldtube. This construction proves smooth source existence, but it does not bound the stress of an autonomous mechanism that generates the prescribed J_n .

For a finite pointer, apply this construction to each conditional datum separately and take a common cutoff interval and worldtube. Because only finitely many branches are present, the approximations can be chosen so that all branch energies and all pairwise thermal forms converge simultaneously. The resulting diagonal controlled source therefore realizes the Schur channel used in theorem 4.1. This observation establishes finite-pointer source closure; it does not add a general- d saturation claim.

6 Relation to prior bounds and detector work

6.1 Observer entropy and the simple random code

Harlow, Usatyuk, and Zhao formulate quantum mechanics in a closed universe using a quantum-to-classical observer channel and test that rule in a simple Haar-random code [1]. Their second-moment expression contains the observer-record factor $e^{-S_2(\omega_{\text{ob}'})} = \text{Tr } \omega_{\text{ob}'}^2$. It is this factor, rather than the binary diamond distance from complete dephasing, that makes contact with the present field model.

The identification in equation (83) is exact because the controlled-displacement record is pure before either subsystem is traced out. The insertion in proposition 4.3 then specializes the matter states so that the two additional overlaps in their formula vanish. Thus our lower floor concerns a Haar-averaged squared inner-product fluctuation for one orthogonal CRT-real pair. We do not infer a deterministic floor for every code realization, a uniform statement over the matter Hilbert space, or a gravitational derivation of the quantum-to-classical channel.

The entropy also has a different role from a thermodynamic or area entropy. $S_2(\rho_P)$ measures the mixedness of the pointer after its conditional field records are discarded; its maximum is the classical collision entropy $H_2(w)$. The branchwise gravity corollary bounds this operational record entropy by a local final-slice backreaction budget. It does not equate the two quantities or assume a Bekenstein–Hawking formula for the pointer.

6.2 Bekenstein entropy is a different quadratic form

For a Klein–Gordon wave packet localized in a spatial region of half-width a , Hollands, Longo, and Morsella prove the Bekenstein-type inequality

$$S(\Phi \mid B) \leq 2\pi a E, \quad (97)$$

where $S(\Phi \mid B)$ is the entropy quadratic form of the local standard subspace [7]. At second quantization this is the relative entropy of the corresponding coherent excitation. This powerful result is directly adjacent to theorem 3.3, but it controls a different functional.

We make the distinction explicit in the simplest common setting. Consider the massless conformal vacuum in four-dimensional Minkowski space, restrict to radial $q = 0$ data in a ball of radius a , and normalize $\|p\|_2 = 1$. The conformal-ball modular Hamiltonian and the half-line dephasing kernel give

$$S_B[p] = \frac{\pi}{2a} \int_0^a (a^2 - x^2) p(x)^2 dx, \quad (98)$$

$$\Gamma_0[p] = \langle p, h_0^{-1} p \rangle = \frac{1}{\pi} \int_0^a \int_0^a p(x) p(y) \log \frac{x+y}{|x-y|} dx dy. \quad (99)$$

Proposition 6.1 (Quadratic-form separation). *At fixed a , there is no finite constant c such that $\Gamma_0[p] \leq cS_B[p]$ for all smooth normalized radial momenta, and no finite constant c' such that $S_B[p] \leq c'\Gamma_0[p]$ for all such data.*

Proof. Let $f \in C_c^\infty(0, 1)$ be positive and normalized in $L^2(0, 1)$. For a packet in a boundary layer, set

$$p_\delta^{\text{wall}}(x) = \delta^{-1/2} f\left(\frac{a-x}{\delta}\right). \quad (100)$$

Substitution in equation (98) gives

$$S_B[p_\delta^{\text{wall}}] = \pi\delta \int_0^1 u f(u)^2 du + O(\delta^2). \quad (101)$$

For the logarithmic kernel, scaling both variables gives

$$\Gamma_0[p_\delta^{\text{wall}}] = \frac{\delta}{\pi} \left(\int_0^1 f(u) du \right)^2 \log \frac{1}{\delta} + O(\delta). \quad (102)$$

Thus $\Gamma_0/S_B \rightarrow \infty$.

For a packet near the center, set

$$p_\delta^{\text{ctr}}(x) = \delta^{-1/2} f(x/\delta). \quad (103)$$

Then

$$S_B[p_\delta^{\text{ctr}}] = \frac{\pi a}{2} + O(\delta^2), \quad \Gamma_0[p_\delta^{\text{ctr}}] = O(\delta), \quad (104)$$

because the logarithmic kernel becomes scale invariant after $x = \delta u$, $y = \delta v$, while the two integration measures leave one factor of δ . Hence $S_B/\Gamma_0 \rightarrow \infty$. $\square$

The separation has a simple physical meaning. The Bekenstein entropy is a local modular-information functional whose conformal weight vanishes at the ball boundary. The zero-temperature dephasing exponent is a global one-particle norm, equivalently an entangling-particle count for the exact detector model. At finite temperature, the factor $\coth(\beta h/2)$ enhances infrared fluctuations further. Thermal dephasing can increase even while the distinguishability of two displaced thermal states decreases. The two notions of information should not be identified.

6.3 Relation to DSW and exact detector work

DSW decoherence describes the loss of coherence of superposed charges or masses near Killing horizons through soft field modes [8, 9]. Gralla and Wei gave a general horizon formulation and precise rates in electromagnetic and Klein–Gordon analogues [10]; Li evaluated the local scalar, electromagnetic, and gravitational cases in de Sitter [11]. These works establish the physical importance of the low-frequency KMS sector that appears in equation (18).

Our setup differs in its question. We do not derive a new horizon decoherence rate. We start from the exact pointer channel and maximize its dephasing functional subject to finite final support and post-switch field energy. The result is consequently a constraint on a local implementation, not a claim that horizon modes require a large Killing energy. Indeed, sufficiently soft horizon radiation can carry negligible Killing energy when its support and duration are allowed to grow. In equation (70), that escape route is visible in the growth of C_{opt} with $L = \ell + T$.

Barcellos and Landulfo already separate field background energy, individual switching work, and communication work for the exact gapless-detector channel [3]. We use their same exact channel class. The new

step is to eliminate the source amplitude, identify the exact compact KMS kernel, and solve the full final-support optimization. We therefore make no novelty claim for Magnus termination, the KMS characteristic function, the coherent stress identity, or switching-energy accounting.

Related Unruh–DeWitt-gate work represents relativistic communication channels as bosonic dephasing channels whose parameters depend on coupling, smearing, and switching data [12]. That work concerns channel structure and capacity rather than the post-switch field-energy and final-support variational problem posed here.

Likewise, spectral extremization of Riesz-potential operators is not new [13], and largest-eigenvalue questions for logarithmic potentials have an established literature [14, 15]. There are two exact relationships that sharpen this comparison. First, let $\tilde{p}$ be the odd extension of p from $(0, 1)$ to $(-1, 1)$. The zero-temperature operator is precisely the odd sector of the finite-interval logarithmic convolution:

$$\frac{1}{\pi} \int_{-1}^1 \log \frac{1}{|u-v|} \tilde{p}(v) dv = \int_0^1 k_0(u, v) p(v) dv, \quad 0 < u < 1. \quad (105)$$

Polosin studies high-index eigenvalue and eigenfunction asymptotics of that convolution in its even and odd sectors [16]. Thus the vacuum convolution problem itself is not new; that work does not give the thermal principal-support law or the uniform temperature remainders used here.

Second, for $\tau > 0$, set $r = e^{-2\tau u}$ and $s = e^{-2\tau v}$. Directly,

$$k_\tau(u, v) = \frac{1}{\pi} \log \frac{1-rs}{|r-s|}. \quad (106)$$

The ratio on the right is the inverse pseudo-hyperbolic distance between two points on a diameter of the Poincaré disk. The change of variables turns $L^2(0, 1)$ into $L^2((e^{-2\tau}, 1), dr/(2\tau r))$. Hence our operator is a weighted one-dimensional geodesic compression of the hyperbolic logarithmic Green kernel, not the two-dimensional hyperbolic-area operator studied in [15].

At the quadratic-form level, odd extension also writes the localized operator $B_{\beta,L}$ as a finite-interval Wiener–Hopf term minus its Hankel image, generated by the singular even multiplier $|k|^{-1} \coth(\beta|k|/2)$. The zero-frequency singularities of the two image terms cancel in their Dirichlet difference. This places the operator near work on singular truncated Wiener–Hopf inverses and determinants [17, 18], although those results concern different spectral quantities and symbol hypotheses.

Thus the claim here is not a first eigenvalue theorem for an unspecified positive compact kernel. It is the identification of the exact reflected thermal operator governing this local pointer channel, proof that it controls the complete canonical and angular phase space, and extraction of uniform support laws with explicit constants. The comparisons above do not by themselves establish priority; they make the remaining operator-theory novelty question precise.

7 Discussion and claim boundary

The result has two layers. The analytic engine is the exact localization coefficient for the thermal inverse Hamiltonian. The physical composition is the finite-pointer purity bound and its insertion into a simple observer code. Together they say that a field record of large second Rényi entropy cannot be produced with simultaneously fixed support and bounded post-switch scalar energy in this model. The statement is quantitative and nonperturbative in the source amplitude.

Compact final support is the decisive hypothesis. Without it, arbitrarily soft thermal modes prevent an energy norm from controlling the dephasing form. With it, the compressed inverse Hamiltonian is bounded. At arbitrary inverse temperature its norm is governed by $\mathcal{K}_{\pi L/\beta}$; in conformal de Sitter, angular resolvent

order and a separate field-coordinate estimate show that the same s-wave momentum mode controls the complete canonical and angular phase space. The temperature–curvature relation enters only in this all-sector reduction.

7.1 What is established

The exact statements are:

1. a controlled-source realization of the finite-pointer Schur channel, with pairwise exponents fixed by conditional field differences;
2. the arbitrary- β coefficient $C_\beta(L) = 2L\Lambda(\pi L/\beta)$, its simple top mode, global bracket, and uniform support asymptotics;
3. the conformal de Sitter reduction of every angular and canonical sector to that unique s-wave momentum optimizer;
4. the purity and Rényi bounds equations (73) and (75), with exact saturation for the symmetric binary top-mode record;
5. the physical-purity insertion equation (85) into the Harlow–Usatyuk–Zhao simple random code for an orthogonal CRT-real matter pair; and
6. the branchwise final-slice constraint estimate equation (121) for spherical $q_i = 0$ records.

The finite-pointer theorem is not claimed to be globally sharp at arbitrary dimension. Simultaneous equality would require every nonzero pairwise difference to align with one simple top eigenspace and all off-diagonal exponents to coincide. The binary construction meets those conditions; a generic larger alphabet does not.

Sharpness of the operator coefficient refers to final Cauchy data supported in $[0, L]$. For a source cylinder with optical radius $\ell < L$ and duration $T = L - \ell$, finite propagation proves the same upper bound but not near-controllability of its top eigenfunction. Smooth compact sources do approach every final-support Rayleigh quotient when their worldtube approaches the declared final-support ball.

7.2 Observer-code interpretation

The code insertion supplies a concrete bridge that the earlier binary diamond-error comparison did not. The conditional field record purifies the physical pointer state, so its record-system purity is exactly $\text{Tr} \rho_P^2$. For the specified orthogonal CRT-real pair, this is precisely the observer factor left in the Haar second moment of [1]. The field theorem therefore converts a localization and energy budget into a lower floor on that ensemble-averaged squared inner-product fluctuation.

This remains narrower than a gravitational observer theorem. The result does not bound the error of every fixed encoding map, establish a high-probability statement, or control all matter pairs at once. It also does not identify the scalar energy with a total observer mass. Harlow’s Hartle–Hawking extension separately argues that the formal $S_{\text{Ob}} \rightarrow \infty$ limit likely requires an infinitely heavy observer [19]. Our result realizes a related energy–entropy obstruction in one field-record sector; it is not a derivation of that broader claim.

The field-energy quantity is the post-switch scalar Killing energy of the coherent conditional displacements. It excludes the work, clock, battery, trap, material support, and stress needed to implement the prescribed sources. The shared finite pointer is a gapless-detector idealization rather than an autonomous relativistic probe field. Accordingly, the theorem is not a universal measurement-energy principle.

The gravity corollary is likewise deliberately branchwise and kinematical. Each conditional spherical datum is placed on its own exact Einstein–scalar final slice and must satisfy the local budget separately. The quantum channel is still evaluated on the unperturbed de Sitter background. We do not derive a common branch-dependent quantum evolution, include actuator stress, or prove global closeness to pure de Sitter; the compact mass shifts the cosmological horizon.

7.3 Novelty boundary

Exact gapless-detector channels, their thermal damping factors, and their field-energy accounting are prior art. Positive compact kernels, simple top eigenvalues, logarithmic-potential operators, and Riesz-potential variational principles are also established mathematics. The candidate contribution is their specific composition here: the reflected thermal operator selected by final support, its exact all-sector de Sitter norm, the finite-pointer Rényi resource bound, and the resulting observer-code mean-square floor.

The localized Klein–Gordon entropy comparison in section 6 shows that this purity functional is not a repackaging of the recent Bekenstein bound. It is not a second headline. Independent review in relativistic detector theory, integral-operator or fractional-Sobolev theory, and quantum gravity remains the binding publication gate.

7.4 Extensions

A first physical strengthening would replace the Haar mean-square statement by a concentration or high-probability theorem over the code ensemble. That requires controlling correlations among matter pairs and is not supplied by the present second moment. A second would replace the prescribed pointer by a localized relativistic probe field, following the detector-from-field framework of [6], while retaining the localization theorem and accounting for the probe stress tensor.

The direct mathematical extension is the true fixed-source-cylinder optimization. It requires either a near-controllability theorem for sources in a prescribed smaller worldtube or a source-to-final-data singular-value problem replacing Λ . A broader geometric extension would identify conditions under which localized norms of $h^{-1} \coth(\beta h/2)$ admit comparable sharp constants on other static spacetimes.

7.5 Reproducibility

From the repository root, the analytic records, independent finite-pointer replay, numerical illustration, and focused checks are reproduced by

```
PYTHONPATH=. python experiments/local_scalar_observer_cost_audit.py
PYTHONPATH=. python experiments/local_scalar_observer_spectrum.py
PYTHONPATH=. python experiments/finite_pointer_observer_audit.py
PYTHONPATH=. python experiments/finite_pointer_observer_clean_room_check.py
PYTHONPATH=. python -m pytest -q \
    tests/test_local_scalar_observer_cost.py \
    tests/test_local_scalar_observer_manuscript.py \
    tests/test_local_scalar_observer_spectrum.py \
    tests/test_finite_pointer_observer.py
```

The frozen records check formulas, source hashes, numerical convergence, and artifact provenance. They do not replace the analytic proofs or certify literature novelty.

A Final-slice Einstein–scalar application

This appendix supplies the backreaction corollary advertised in the title; it is not a coupled-gravity extension of the channel theorem. The all-angular de Sitter result does not require spherical symmetry. We now restrict to the s-wave and to a final slice on which the radial field coordinate vanishes. This subclass is useful because its momentum density vanishes and its energy measure has a direct constraint interpretation.

Let $u(t, x)$ be the normalized rescaled s-wave. The physical conformal scalar is

$$\phi(t, r) = \frac{u(t, x(r))}{\sqrt{4\pi} r}. \quad (107)$$

On a final slice with $u = 0$ and $p = \partial_t u$, the physical normal derivative in the de Sitter geometry is

$$v(r) := n^a \nabla_a \phi = \frac{p(x(r))}{\sqrt{4\pi} r \sqrt{N(r)}}. \quad (108)$$

For the conformally coupled stress tensor, every improvement term containing ϕ or a spatial derivative of ϕ vanishes on this slice. Directly,

$$\rho = T_{ab} n^a n^b = \frac{1}{2} v^2, \quad j_i = -T_{ab} n^a h^b_i = 0. \quad (109)$$

Since $dr = N dx$, equations (108) and (109) imply the exact measure identity

$$4\pi r^2 \rho(r) dr = \frac{1}{2} p(x)^2 dx. \quad (110)$$

Use the same numerical function v as the scalar normal-momentum datum on a new spherical initial slice, and consider gravitational data with vanishing extrinsic curvature,

$$dl^2 = \frac{dr^2}{f(r)} + r^2 d\Omega_2^2, \quad K_{ij} = 0, \quad f(r) = N(r) - \frac{2Gm(r)}{r}. \quad (111)$$

The momentum constraint is satisfied by equation (109). The Hamiltonian constraint with cosmological radius R is solved by

$$m'(r) = 4\pi r^2 \rho(r), \quad m(0) = 0. \quad (112)$$

Absorbing the Bunch–Davies vacuum expectation into the background cosmological constant, the coherent excess stress in equation (29) supplies precisely the classical source above. Although $K_{ij} = 0$, the nonzero scalar normal momentum means that the full Einstein–scalar data are not invariant under time reflection. Equation (110) then gives

$$m(b) = E_K \quad (113)$$

for any areal radius b outside the field support.

Define the dimensionless radial constraint ratio

$$Q(r) = \frac{2Gm(r)}{rN(r)}. \quad (114)$$

For a comparison ball $0 < r \leq b < R$, set

$$Q_b := \sup_{0 < r \leq b} Q(r). \quad (115)$$

If $Q_b \leq \delta < 1$, then throughout that ball

$$f(r) \geq N(r)(1 - \delta) > 0, \quad 0 \leq \frac{g_{rr}}{g_{rr}^{\text{dS}}} - 1 \leq \frac{\delta}{1 - \delta}. \quad (116)$$

Corollary A.1 (Necessary final-slice constraint cost). *Let the hypotheses of corollary 3.6 hold and suppose the source is engineered to finish with spherical $q = 0$ data. Set*

$$b = R \tanh(L/R), \quad L = R \operatorname{arctanh}(a/R) + T. \quad (117)$$

Then

$$Q_b \geq Q(b) \geq \frac{2G}{bN(b)RC_{\text{opt}}(L/R)} \log \frac{1}{2\epsilon_{\text{obs}}}. \quad (118)$$

Equivalently, a local budget $Q_b \leq \delta < 1$ implies

$$\epsilon_{\text{obs}} \geq \frac{1}{2} \exp \left[-\frac{\delta b N(b) R}{2G} C_{\text{opt}}(L/R) \right]. \quad (119)$$

Proof. Outside the support, equation (113) gives $Q(b) = 2GE_K/[bN(b)]$. Apply the sharp energy lower bound in corollary 3.6; solving the same inequality in the other direction gives equation (119). $\square$

The binary statement has a finite-pointer counterpart only when the constraint budget holds on every conditional branch. A bound on the weighted mean field is insufficient because different branches may carry large and opposite masses while their mean vanishes.

Corollary A.2 (Branchwise constraint bound on pointer entropy). *Let the hypotheses of corollary 4.2 hold with spherical conditional data $q_i = 0$ supported in $[0, L]$. Set*

$$y = L/R, \quad b = R \tanh y, \quad N(b) = \operatorname{sech}^2 y. \quad (120)$$

For each branch, let $Q_{b,i}$ be the local constraint ratio defined by equation (114) for that branch. If $Q_{b,i} \leq \delta < 1$ for every i , then

$$S_2(\rho_P) \leq \delta \frac{R^2}{G} \frac{C_{\text{opt}}(y) \tanh y \operatorname{sech}^2 y}{2}. \quad (121)$$

Proof. For branch i , the exact mass–energy identity gives $E_i = m_i(b)$ outside the support. Since $Q_i(b) \leq Q_{b,i} \leq \delta$,

$$E_i \leq \frac{\delta b N(b)}{2G}. \quad (122)$$

The centered energy obeys

$$\bar{E}_K = \sum_i w_i E_i - E[\bar{q}, \bar{p}] \leq \sum_i w_i E_i \leq \max_i E_i. \quad (123)$$

Combining this inequality with equation (81) and using $b = R \tanh y$ proves equation (121). $\square$

At $y = 1$, the converged Galerkin estimate $C_{\text{opt}}(1) = 1.0295979905445$ gives the illustrative coefficient

$$S_2(\rho_P) \lesssim 0.1646584608127 \delta \frac{R^2}{G}. \quad (124)$$

This decimal is not a certified value of the continuum eigenvalue. Using the rigorous row-Schur estimate

$$C_{\text{opt}}(1) \leq 1.1594713304692 \quad (125)$$

instead gives

$$S_2(\rho_P) \leq 0.1854284549745 \delta \frac{R^2}{G}. \quad (126)$$

The exact analytic statement is equation (121); the numbers only illustrate its scale.

The compact mass shifts the cosmological horizon relative to the pure de Sitter value R . Accordingly, Q_b is a local comparison norm on the ball containing the final datum; no global smallness claim relative to the unperturbed lapse is intended.

A.1 An explicit local weak-constraint window

For the source optical radius $\ell = R \operatorname{arctanh}(a/R)$, take the ideal profile

$$q = 0, \quad p(x) = A \sqrt{\frac{3}{\ell^3}} x \mathbf{1}_{(0,\ell)}(x). \quad (127)$$

Its cumulative mass is

$$m(x) = E_K \left(\frac{x}{\ell} \right)^3. \quad (128)$$

Inside the support, the function

$$\frac{\operatorname{arctanh}(r/R)^3}{r[1 - r^2/R^2]} \quad (129)$$

is strictly increasing on $(0, a]$. One way to see this is to factor it as $(x/r)(x^2/N)$; both factors increase with r . Hence the wall value controls the complete profile. Combining equation (92) with the wall constraint gives the sufficient estimate

$$Q_a \leq \frac{2\pi G \Gamma_\star}{3\ell a[1 - a^2/R^2]}. \quad (130)$$

For illustration, set

$$\epsilon_{\text{obs}} = 10^{-6}, \quad a/R = 0.2, \quad T/R = 0.1, \quad G/R^2 = 10^{-6}, \quad \delta = 0.25. \quad (131)$$

The universal theorem requires

$$C_{\text{opt}}(L/R) \leq 0.340766570374, \quad E_K R \geq 38.5083647231, \quad Q(b) \geq 2.86896268303 \times 10^{-4}, \quad (132)$$

whereas the constructive family has

$$E_K R \leq 67.7824380936, \quad Q_a \leq 7.06067063475 \times 10^{-4}. \quad (133)$$

Thus the fixed numerical point has a nonempty local weak-constraint window. By lemma 5.1, smooth compact profiles preserve this strict margin.

The construction closes only the final Hamiltonian and momentum constraints. It does not specify the lapse, evolve the Einstein–scalar system, include the source actuator during switching, or recompute the quantum channel in the perturbed geometry.

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